GLOBAL CREDIT TRADER

AI, Credit, and a Higher Cost of Capital

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- Higher rates have led to increased questions around the funding requirements for AI capex, leading to a correction across some risk assets. In credit, the impact has been most pronounced in USD HY. While some softer AI-related sentiment can move datacenter bonds wider, we view AI-related corporate bonds from neoclouds or those with frontier lab exposure as being at the epicenter of the funding cost debate.
- The intersection of existing fundamentals and this reassessment of funding costs and returns means that the sell-off resembles the broader bifurcation in USD HY: index-level spreads remain relatively supported, but dispersion is elevated.
- Our equity research colleagues expect AI-related capex to remain elevated in the near term as demand continues to outstrip supply. We remain underweight USD IG Technology. In USD HY, we remain neutral, as we see scope for some near-term reversal of the recent sell-off and valuations are more attractive, especially given the structural nuances for project finance bonds. However, dispersion across AI-related issuers is likely to remain high.
- More debates around AI funding costs increase focus on the growing project finance issuance for AI capex. USD IG and HY project finance issuance year to date has surpassed $106 billion, nearly triple the $37 billion in 2025.
- The pace of issuance and magnitude of the financing need leaves risks to our 2027 forecast of $300 billion, globally, skewed firmly to the upside.
- Performance across HY datacenter bonds has been challenged recently, with average dollar prices falling from above $101 in June to nearly $95 today. However, the price action has been dispersed. As a result, we continue to advocate for patient, project-specific underwriting for these names.
- As the pace of issuance has increased, differentiation factors around construction risk, coverage of principal or refinancing risk, and claim to tenant have become increasingly important. Additional considerations around power availability, permitting, and insurance all advocate for a bottom-up approach to investing in the asset class.

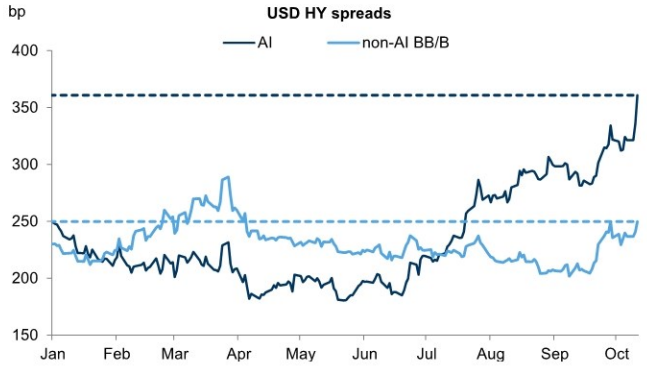
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AI, Credit, and a Higher Cost of Capital

Questions around the funding requirements of the AI capex buildout have become more pressing, leading to a correction across risk assets. In credit, the impact has been most pronounced in USD HY. USD HY AI-related spreads widened by 25bp between October 13 and 14, reaching a new high of 360bp over Treasuries (Exhibit 1). By contrast, non-AI-related spreads were only 2bp wider and remain below their March peak.

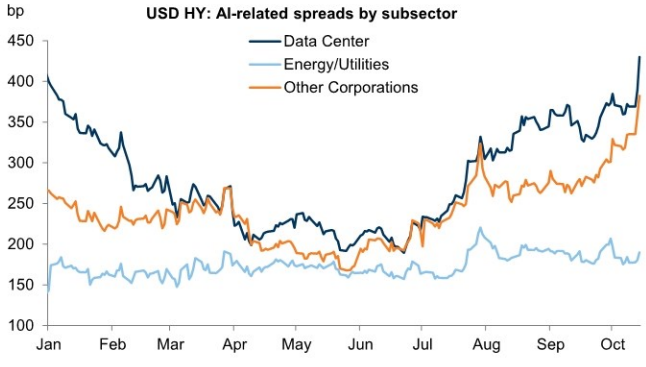
Within AI-related issuers, the widening has been concentrated in AI-related corporate bonds and data center project finance (Exhibit 2). While some softer AI-related sentiment can move datacenter bonds wider, especially if there is a reassessment around off-taker credit risk, we view AI-related corporate bonds, from either neoclouds or firms with significant frontier lab exposure, as being at the epicenter of the funding cost debate. Further removed AI-adjacent Energy and Utilities have been more resilient, in our view partly reflecting more diversified revenue streams and a relatively defensive earnings profile.

Exhibit 1: USD HY AI-related spreads have widened significantly



Source: iBoxx, Goldman Sachs Global Investment Research

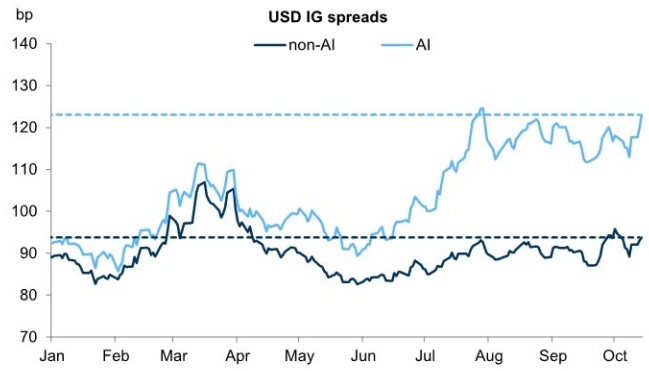
Exhibit 2: Data centers and related issuers have driven the widening



Source: iBoxx, Goldman Sachs Global Investment Research

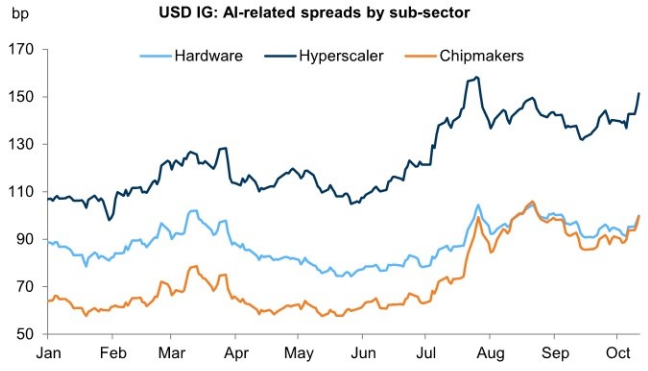
The price action has been less severe in IG, reflecting generally stronger financial positions for the higher-rated cohort. USD IG AI-related spreads are only 3bp wider vs. 1bp for non-AI-related issuers (Exhibit 3). Spreads have moved wider across both hardware and Hyperscalers, although funding concerns have driven a decompression bias across and within these cohorts (Exhibit 4).

Exhibit 3: USD IG AI-related spreads have widened but remain below summer peaks



Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 4: For IG AI issuers, both hardware and Hyperscalers have moved wider



Source: iBoxx, Goldman Sachs Global Investment Research

We think the widening reflects a reassessment of the economics of the AI buildout, particularly for lower-rated borrowers that are more exposed to higher-for-longer rates and potential energy-cost pressures. In this respect, the sell-off resembles the broader bifurcation already evident in USD HY: index-level spreads remain relatively tight, but dispersion is elevated.

More broadly, our equity analysts expect AI-related capex to remain elevated in the near term, as demand continues to outstrip supply. We therefore remain underweight USD IG Technology. In USD HY, we see scope for some near-term reversal of the recent sell-off and remain neutral given more attractive valuations. However, dispersion across AI-related issuers is likely to remain high.

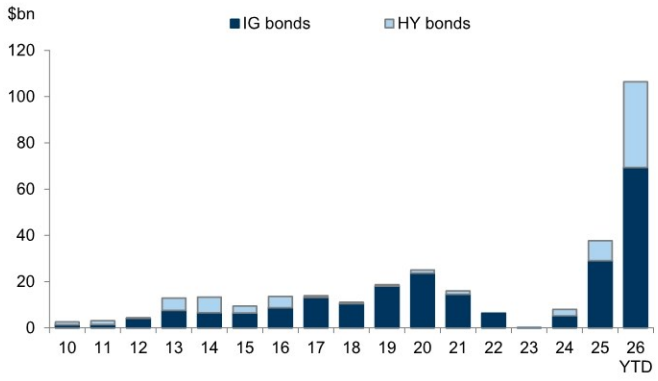
Project finance issuance—a paradigm shift for credit markets

As the cost of funding debate continues, we think this puts more of a spotlight on the project finance ecosystem that has been tapped to help fund this unprecedented, multi-year buildout.

As shown in Exhibit 5, in the USD IG and HY bond markets, project finance issuance has already surpassed $106 billion, which is nearly three times the full-year 2025 level of $37 billion.

For 2027, we are expecting $300 billion of global project finance issuance across the IG and HY markets to fund data center and compute investments. This is in addition to the $420 billion of direct issuance from the hyperscalers that we forecast for next year. Given the magnitude of the financing need, we view the risks to our $300 billion project finance projection as firmly skewed to the upside. While a higher cost of capital may shift the timeline somewhat, we see a high bar to derailing the investment.

Exhibit 5: Project finance issuance in the USD IG and HY bond markets has already surpassed $106 billion—nearly 3x the full-year 2025 level of $37 billion
USD IG and HY project finance gross issuance volumes ($bn)



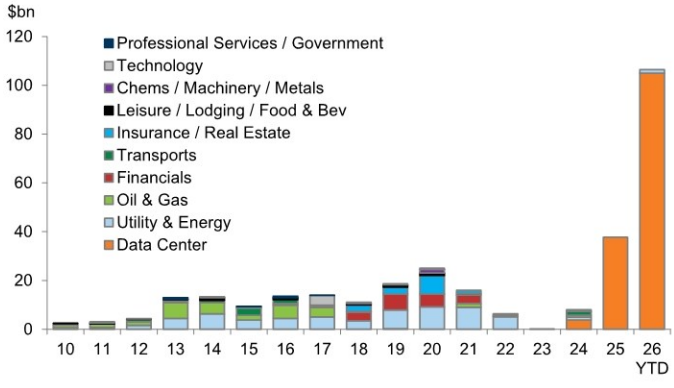
Note: As of October 13, 2026.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

For much of the post financial crisis era—and until recently—project finance issuance was dominated by the investment grade market (IG), in sectors such as Utilities and Financials. In 2017 through 2021, for example, much of this supply was associated with sustainable financing projects. This included green bonds issued for renewable energy and energy efficiency as well as social bonds for affordable housing.

This sector composition shifted, however, beginning in 2024 as issuance for data center financing accelerated, with more involvement from the high yield (HY) universe. As project finance volumes increased further in 2026, chip financings became more common (which we capture in the Data Center category, Exhibit 6).

Exhibit 6: The recent project finance issuance in the USD IG and HY bond markets has been dominated by data centers
USD IG and HY project finance gross issuance ($bn), by sector



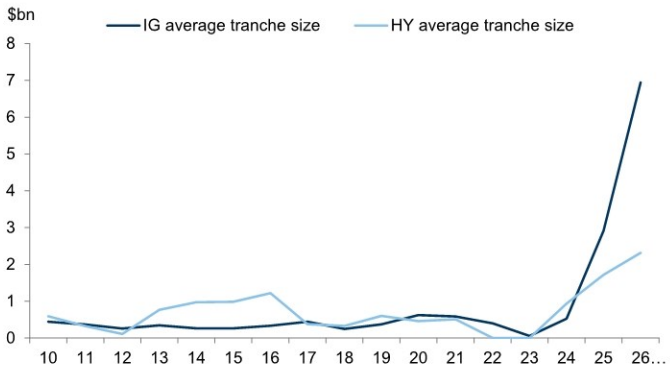
Note: As of October 13, 2026. The Data Center label includes Chip / Compute financing, the volume of which increased in 2026.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

As the data-center-related supply accelerated, the average tranche size has grown substantially. For example, the average tranche size in the USD IG market for project finance issuance is $6.9 billion so far in 2026. In the USD HY market, it is $2.3 billion

(Exhibit 7).

Exhibit 7: The average tranche size for the recent project finance issuance has grown meaningfully as the data center financing has accelerated
USD IG and HY project finance average tranche size, by year ($bn)



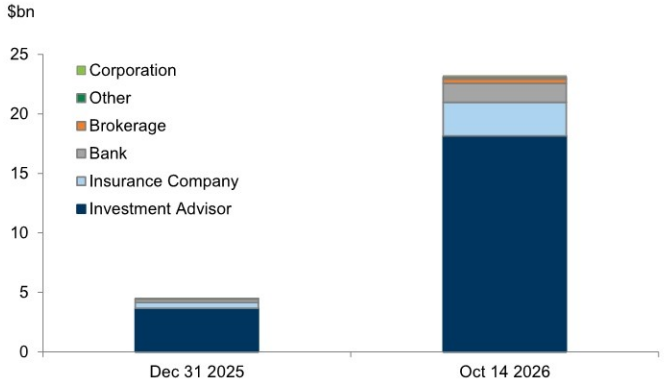
Note: As of October 13, 2026.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

One key question that we commonly receive is which investor types are purchasing this debt. To answer this question, we focus on the USD HY project finance universe, where the pace of issuance – and number of issuers – has been greater vs. USD IG.

Exhibit 8 aggregates reported holdings (as tracked by Bloomberg data) for 20 USD HY project finance bonds, with a total notional outstanding of $45.6 billion. As of October 14, the most recent holdings data mapped $23.3 billion of holdings (as some investors' holdings are not reported). Investment advisors represent the largest category of holders, with $18.2 billion of aggregated holdings (vs. $3.7 billion as of year-end 2025). Insurance companies hold another $2.8 billion (vs. $0.5 billion at year-end 2025).

Exhibit 8: Investment advisors and insurance companies represent the largest holder types of USD HY project finance bonds
Holdings ($bn) by investor type for a universe of 20 USD HY project finance bonds



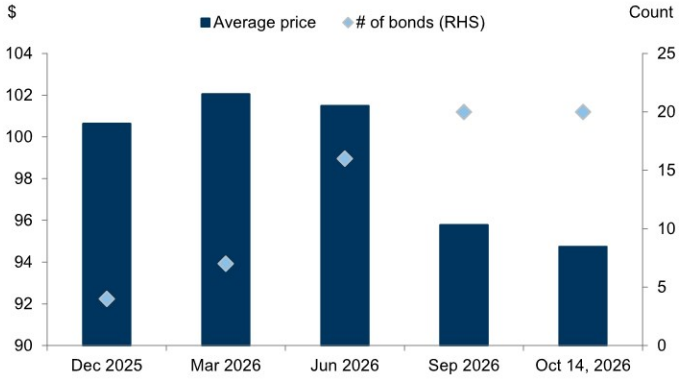
Source: Bloomberg, Goldman Sachs Global Investment Research

The performance of these bonds has been challenged over the course of the last few months, in particular. In Exhibit 9, we track this same universe of 20 bonds and track how the average price of the universe has evolved alongside the issuance. The average

price of the bonds dropped to $94.7 as of October 14th, from $101.5 as of June 30th. That said, there is dispersion. Many of the 2025 and early 2026 vintages are still trading close to (or above) par, given a combination of higher coupons, and near-term data hall delivery. However, more recent issues are trading closer to the low-to-mid $90s.

Rates have also played a meaningful role in the price pressure of the subsector. Tracking the average price of the entire Bloomberg USD HY Corporate Index shows a roughly $4 point drop vs. June 30th: from $97.1 to $93.4 (as of October 14).

Exhibit 9: As interest rates have risen in 2H2026, the dollar price of USD HY project finance bonds has declined
Average prices for the universe of USD HY data center bonds, and the number of bonds in our sample at each point in time



Note: As of October 14, 2026.

Source: Bloomberg, Goldman Sachs Global Investment Research

Key characteristics to monitor

Earlier this year, we outlined some of the key aspects of the data center project financing transactions, highlighting the "Three Cs" which we viewed to be most critical to their analysis. Those were:

1. Construction risk: How far along, who bears cost overruns, and are there lease termination rights if construction is delayed?
2. Coverage: Is there refinancing risk, or will the deal fully amortize within the lease term?
3. Claim to hyperscaler: What is the structure of the hyperscaler involvement (if any)? Are lease payments guaranteed? And are any other operating costs (electricity, taxes, insurance) guaranteed by the lease tenant?

As issuance has continued over the past several months, these differentiating features have become even more important drivers of the dispersion in the subsector. Moreover, it has become clear that other key factors are top of mind for investors, including power availability, permitting, insurance coverage, and location. For example, power availability is emerging as a key constraint for AI infrastructure, and grid connection queues in the US and Europe are set to rise over the next five years. Moreover, constraints in the insurance market could result in higher risk retentions and lower loss limits.

This underpins our call for a specific, deal-by-deal approach to the sector. These bonds should be underwritten on project specifics, some of which are outlined above. In our

view, while there will be some AI-related macro beta for these structures, especially around the left tail, layered with the technical overhang we would be cautious of a broad, top-down driven move meaningfully tighter.

The authors would like to thank Patricia Pacheco for her contribution to this report.

Forecast dashboard: Spreads, returns, issuance, defaults, and ratings migrations

Exhibit 10: USD and EUR spread forecasts

Market	Updated as of Oct. 14, 2026			
	Current	2026Q4	2027Q1	2027Q2
USD spreads (bp)				
USD IG	83	82	80	78
USD HY	324	295	290	280
EUR spreads (bp)				
EUR IG	94	88	86	84
EUR HY	352	315	310	305

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 11: USD and EUR excess returns forecast

Market	Updated as of Oct. 14, 2026		
	Current duration	2026 YTD (%)	2026 Full-year forecast (%)
USD excess returns			
USD IG	6.6	0.4%	0.6%
USD HY	3.6	0.1%	1.3%
EUR excess returns			
EUR IG	4.5	0.3%	0.7%
EUR HY	3.3	-0.5%	1.2%

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 12: Issuance forecasts

Market	Updated as of Oct. 14, 2026		
	2026 YTD (bn)	2026 Full-year forecast (bn)	2027 Full-year forecast (bn)
USD market			
Gross issuance			
IG	$1,904	$2,300	$2,400
HY	$283	$340	$375
Leveraged loans	$348	$400	$500
CLOs	$330	$405	$360
Net issuance			
IG	$787	$1,000	$1,150
HY	$96	$120	$115
Leveraged loans	$32	$50	$75
EUR market			
Gross issuance			
IG	€ 690	€ 850	€ 900
HY	€ 88	€ 115	€ 130
Leveraged loans	€ 66	€ 75	€ 65
CLOs	€ 100	€ 130	€ 125
Net issuance			
IG	€ 177	€ 265	€ 310
HY	€ 14	€ 30	€ 35
Leveraged loans	€ 26	€ 25	€ 20

Source: Dealogic, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 13: Ratings migration and default forecasts
Default rates are issuer-weighted and 12-month trailing

Market	Updated as of Oct. 14, 2026		
	2026 YTD	2026 Full-year forecast	2027 Full-year forecast
USD market			
Rising stars (bn)	$78	$75	--
Fallen angels (bn)	$38	$60	--
HY defaults (12-month trailing)	3.4%	4.0%	4.0%
Leveraged loan defaults (12-month trailing)	4.9%	4.75%	5.0%
EUR market			
Rising stars (bn)	€ 22	€ 20	--
Fallen angels (bn)	€ 5	€ 30	--
HY defaults (12-month trailing)	4.7%	5.0%	5.0%
Leveraged loan defaults (12-month trailing)	3.5%	4.0%	4.0%

Source: Bloomberg, Moody's, Goldman Sachs Global Investment Research

Forecast dashboard: Relative value views

Exhibit 14: Relative value views

Relative value views		
Corporate Credit		
Regional	Overweight USD vs. EUR	We have a slight preference for USD credit vs. its EUR peer, owing to a somewhat more challenging growth-inflation-monetary policy mix in the Euro Area. That said, the lines are blurring between the regional classifications, given the acceleration in cross-border issuance.
	Overweight EUR IG CLO tranches vs. USD IG CLO tranches	EUR IG tranches offer carry pickup relative to USD IG CLO tranches. While the policy and macroeconomic picture is less supportive in Europe, subordination should protect tranches up in the capital structure from a continued idiosyncratic default picture.
USD market	Neutral IG vs. HY	While we previously had a slight preference for IG vs. HY, we now view the case for an outright overweight in IG as less obvious based on valuations, rate sensitivity and supply related technicals.
	Overweight BBBs vs. higher-rated cohorts	We prefer owning BBBs over the higher-rated segments of USD IG (As and AAs), as a way to capture additional spread. We also see scope for balance sheet deterioration at the highest end of the rating spectrum in IG.
	Overweight Bs, Neutral BBs, Underweight CCCs	We expect BBs will continue to encounter a challenging supply technical due to AI-related supply. As a result, we favor the B rated cohort. The CCC universe is highly idiosyncratic and requires careful credit selection, in our view.
	Overweight HY bonds vs. loans	The substantial software sector exposure in the loan market leaves us favoring the HY market, at least until we receive more clarity on the refinancing of the 2028 loan maturity walls. The lower-rated distribution of loans and the limited carry 'pick-up' vs. HY are also contributing factors.
	Overweight CLO mezz tranches vs. leveraged loans Neutral CLO vs. USD IG Neutral CLO vs. CMOs	For floating rate investors, we prefer CLO mezzanine tranches vs. leveraged loans. Historical data points to low impairment rates for BB and BBB CLO tranches, and current valuations offer attractive carry and a cushion against moderate to elevated default outcomes. Elsewhere, we are neutral between senior CLO tranches and USD IG; we would use spread widening to add exposure in CLOs. We are also neutral between CLO AAA and CMO floaters.
	Prefer Agency MBS vs. IG	Agency MBS offers an attractive diversification benefit vs. a portfolio of IG corporate bonds.
	Overweight Hybrids vs. BBBs	We view the additional spread compensation of hybrids vs. BBBs as moderately attractive in the context of the tight spread environment. We also expect the hybrid spread premium to compress vs. BBBs, as the hybrid market grows further.
EUR market	Neutral IG vs. HY	While we previously had a preference for IG vs. HY, we now view the case for an outright overweight in IG as less obvious based on valuations, rate sensitivity and supply related technicals.
	Overweight BBBs vs. higher-rated cohorts	We recently shifted our view to favor BBBs over both AAs and As in the EUR market, owing to an improved macroeconomic backdrop which now favors a broad-based, down-in-quality stance within IG.
	Overweight HY bonds vs. loans	Similar to the US, we prefer HY bonds given the loan market's weighting toward software (and the potential negative sentiment overhang). The reaction function of the ECB has increased borrowing costs for floating rate issuers, given the central bank's focus on inflationary pressures.
	Overweight BBs, Neutral Bs, Underweight CCCs	At current valuations and given our expectation for heightened dispersion, we prefer moving up in quality within EUR HY.

Source: Goldman Sachs Global Investment Research

Performance dashboard: Cash and synthetic markets

Exhibit 15: Performance of USD and EUR markets

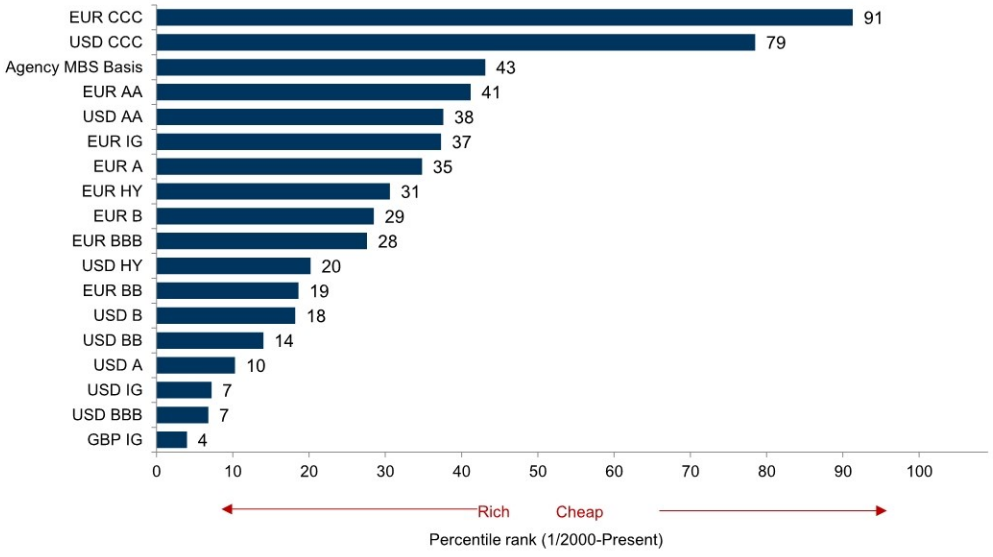
Market	Current*		Total return			Spread change (YTD,bp)	Yield change (YTD, %)
	Spread	Yield	MTD	QTD	YTD		
USD Corporate bonds							
IG	83	6.0	0.73	0.73	-2.41	5	1.17
AA	60	5.8	0.73	0.73	-3.21	11	1.16
A	70	5.8	0.78	0.78	-2.12	5	1.09
BBB	104	6.2	0.62	0.62	-1.80	3	1.10
HY	324	8.4	-0.16	-0.16	-0.06	58	1.88
BB	199	7.2	0.07	0.07	0.02	30	1.28
B	329	8.7	-0.31	-0.31	0.81	28	1.30
CCC	1282	17.8	-2.10	-2.10	-5.47	397	4.83
USD Leveraged Loans							
Leveraged loan index	439	8.5	-0.21	-0.21	3.07	44	0.6
BB	263	6.5	-0.12	-0.12	4.04	7	0.2
B	420	8.2	-0.12	-0.12	3.49	28	0.4
CCC	2306	28.8	-1.86	-1.86	-7.72	742	8.2
USD Securitized products							
Agency MBS	38	5.8	1.05	1.05	-2.29	16	1.2
ABS (Autos)	62	5.3	0.14	0.14	1.94	-4	1.1
ABS (Credit cards)	33	5.1	0.15	0.15	1.17	0	1.2
ABS (Student loans)	94	5.1	0.06	0.06	3.73	121	0.2
USD Macro indices							
CDX IG	59	--	0.00	0.00	0.47	9	--
CDX HY	340	--	-0.27	-0.27	0.98	24	--
EUR Corporate bonds							
IG	106	4.4	0.49	0.49	-0.77	16	0.9
AA	77	4.2	0.56	0.56	-1.12	15	0.8
A	97	4.3	0.52	0.52	-0.77	13	0.9
BBB	122	4.5	0.43	0.43	-0.71	20	1.0
HY	326	6.8	-0.55	-0.55	-0.48	56	1.2
AT1	309	6.3	0.33	0.33	1.54	23	1.0
EUR Leveraged loans							
Leveraged loan index	480	7.6	-0.94	-0.94	2.01	30	1.0
BB	279	5.4	-0.35	-0.35	3.56	-73	-0.1
B	448	7.2	-0.93	-0.93	2.16	33	1.0
EUR Macro indices							
iTraxx Main	64	--	0.01	0.01	0.4	13	--
iTraxx Xover	314	--	-0.44	-0.44	2.6	70	--
iTraxx SenFin	72	--	-0.06	-0.06	0.4	17	--
iTraxx SubFin	120	--	-0.18	-0.18	0.9	27	--

*Data as of previous day close.
For USD IG and HY we use the Bloomberg indices with history back to Jun-89 and Jan-94, respectively. Corp rating level, Agency MBS and ABS we use ICE-BAML indices with history back to Dec-96, EUR IG we use the iBoxx QW5A index with history back to Jan-99, EUR HY we use the ICE-BAML index with history back to Dec-97, and USD and EUR leveraged loans we use the Morningstar LSTA LLI and ELLI indices with data going back to Jan-97 and Jan-02, respectively.

Source: Bloomberg, iBoxx, ICE-BAML, Pitchbook LCD, Goldman Sachs Global Investment Research

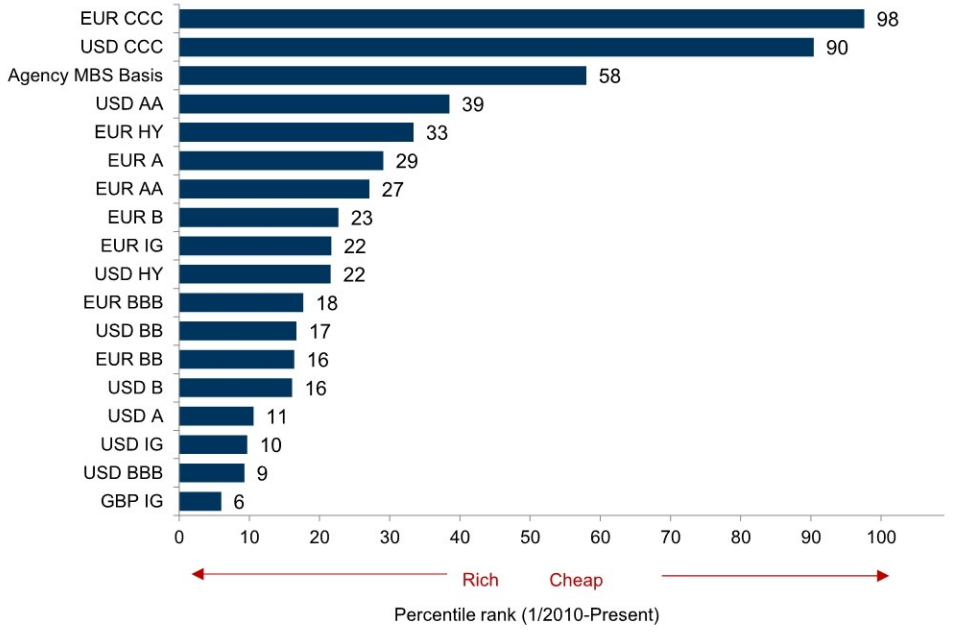
Valuation dashboard: Percentile ranks for spreads and yields

Exhibit 16: Percentile ranks for spreads across markets (2000-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

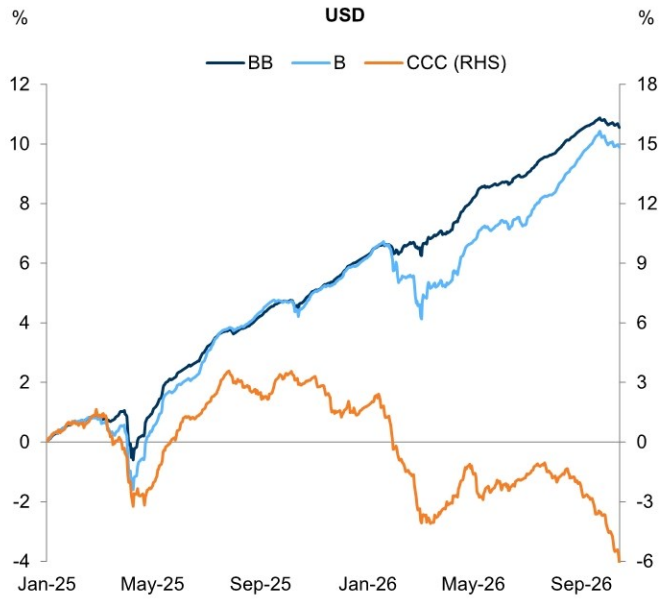
Exhibit 17: Percentile ranks for spreads across markets (2010-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

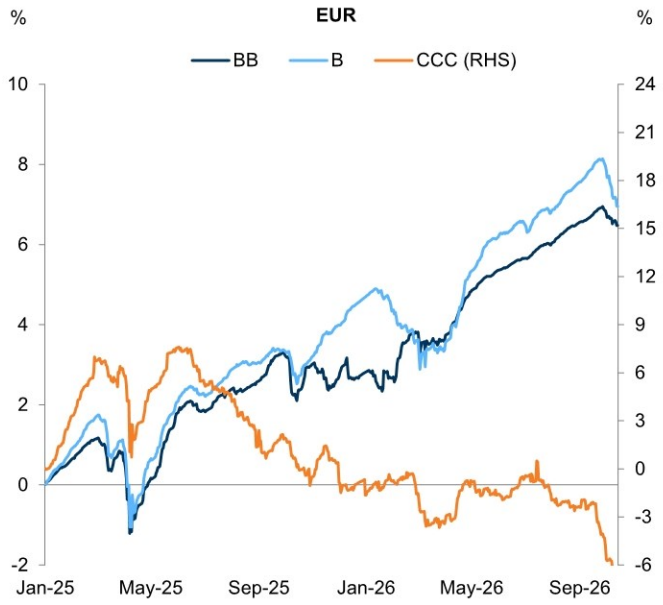
Performance dashboard: Leveraged Loans

Exhibit 18: Cumulative total return by rating for USD Leveraged Loans



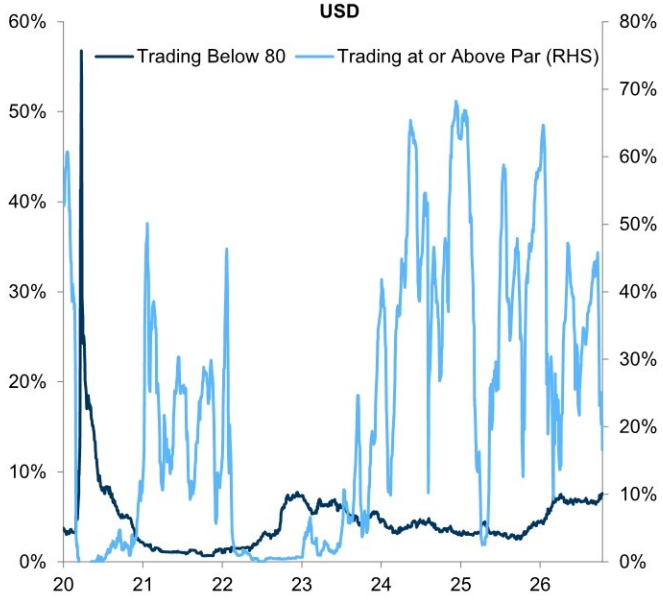
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 19: Cumulative total return by rating for EUR Leveraged Loans



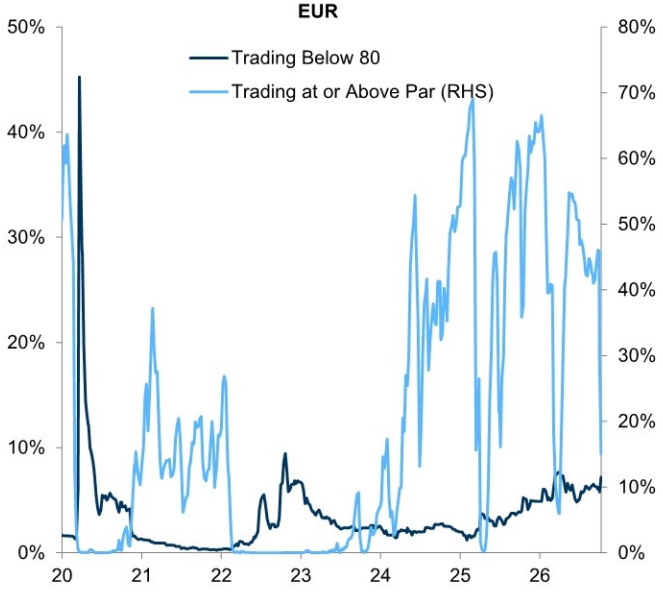
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 20: Price breakdown for USD Leveraged Loans



Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 21: Price breakdown for EUR Leveraged Loans

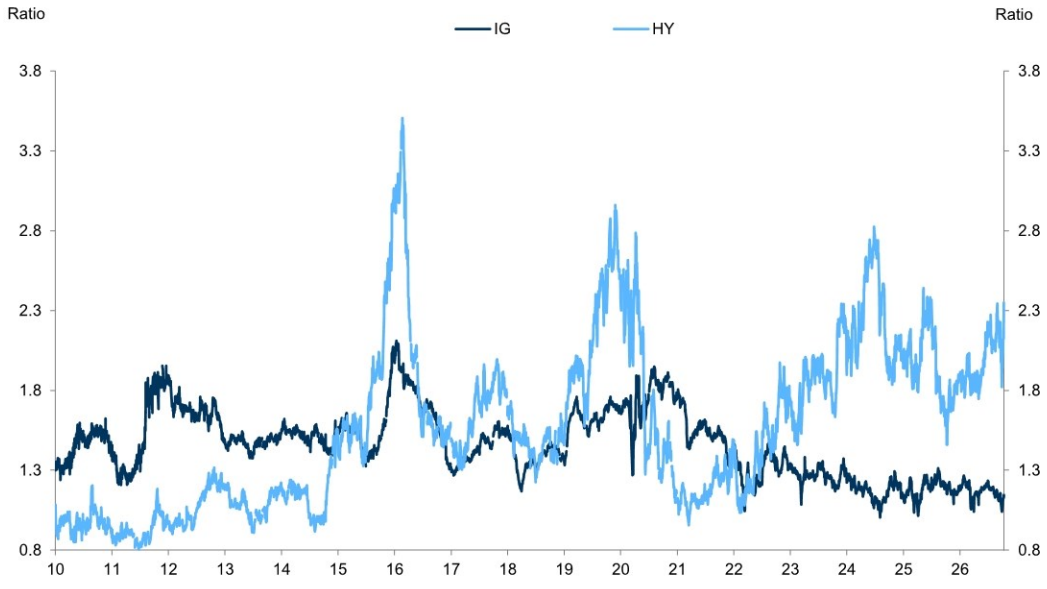


Source: PitchBook, Goldman Sachs Global Investment Research

Performance dashboard: Bond-level dispersion

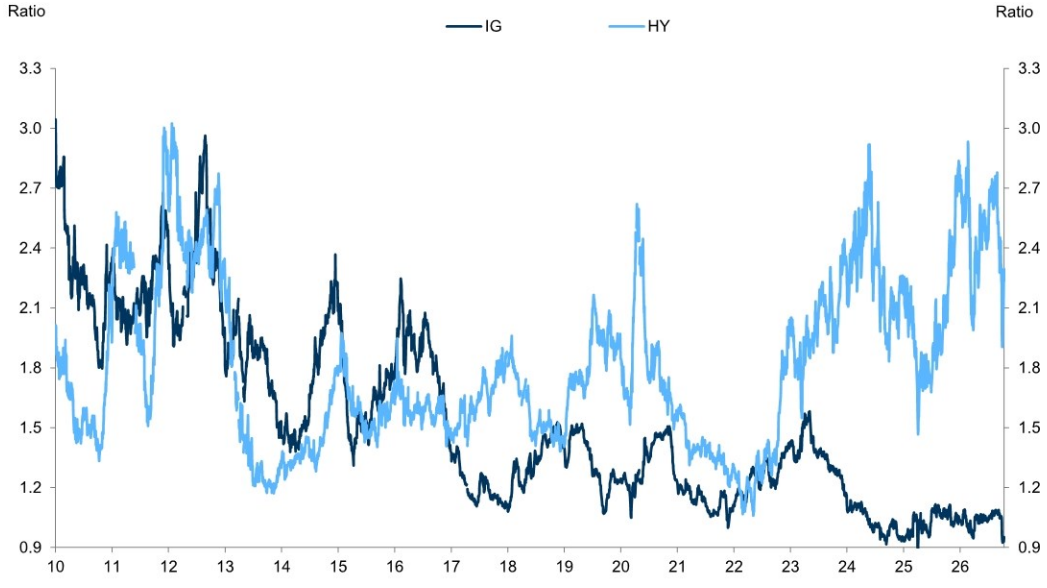
Note: Spread dispersion is the inter-decile range divided by the median across the bond index constituents and return dispersion is the inter-decile range of 10-day returns divided by the median across the bond index constituents.

Exhibit 22: USD spread dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

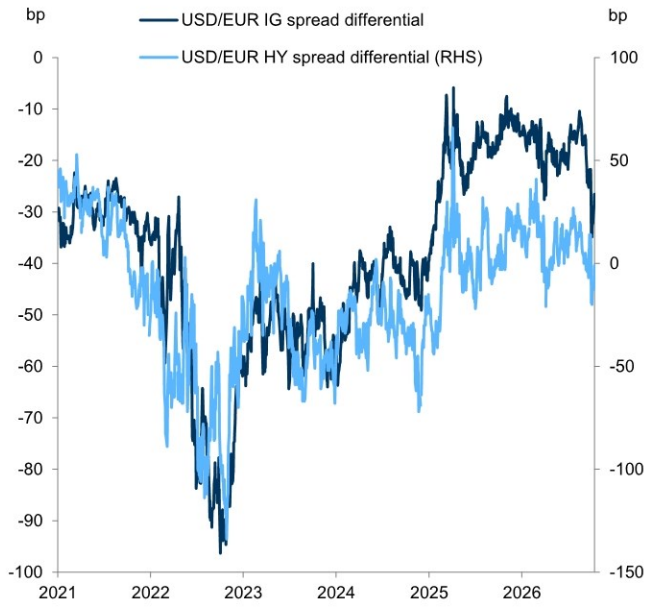
Exhibit 23: EUR spread dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

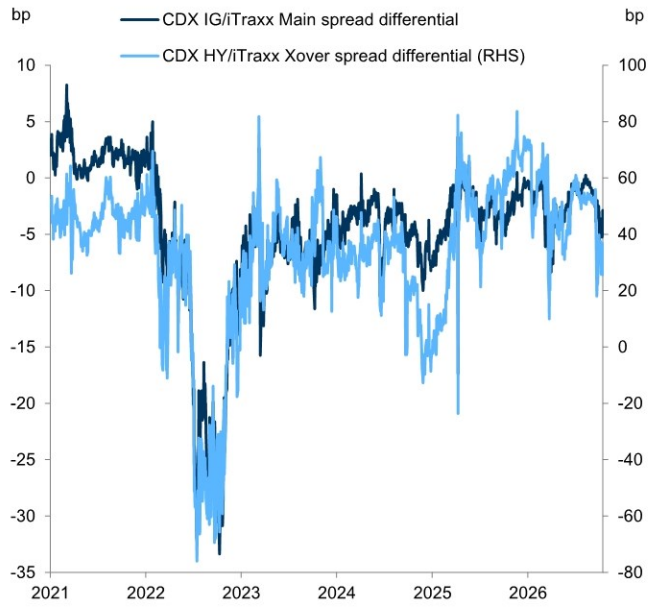
Relative value dashboard: Excess spreads in the USD and EUR markets

Exhibit 24: USD/EUR IG and HY spread differential



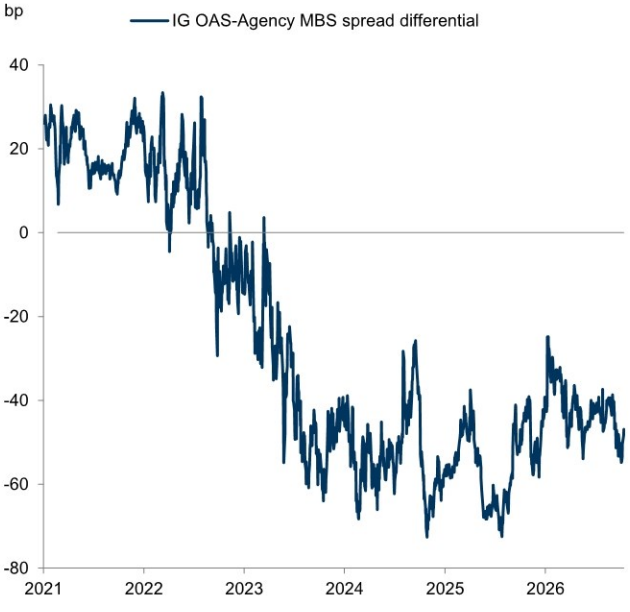
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 25: CDX IG/iTraxx Main and CDX HY/iTraxx Xover spread differential



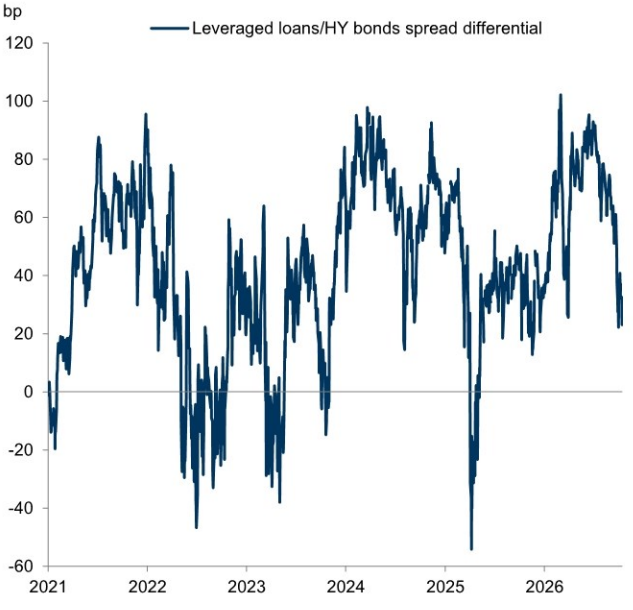
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 26: IG OAS/Agency MBS spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 27: Leveraged loans/HY spread differential



Source: Bloomberg, Morningstar LCD, Goldman Sachs Global Investment Research

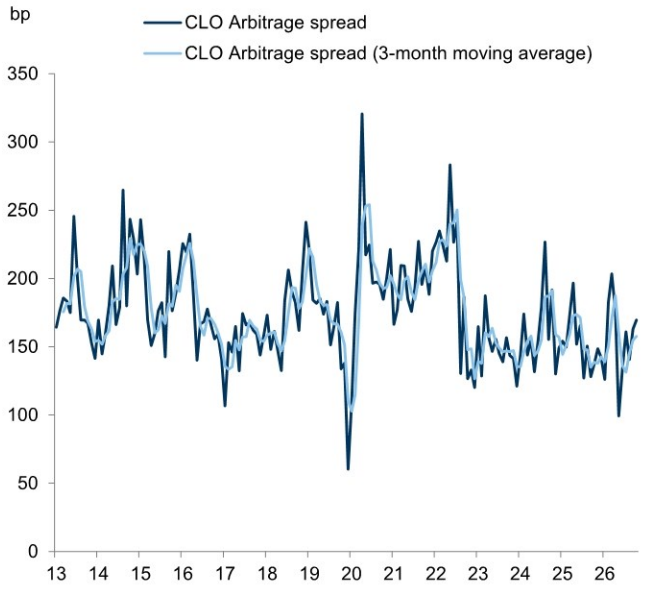
Valuation dashboard: CLOs

Exhibit 28: CLO spread percentile ranks

USD	Current spread (bp)	Percentile rank		
		3-year	5-year	10-year
AAA	117	49%	31%	20%
AA	148	24%	15%	8%
A	175	30%	18%	9%
BBB	287	43%	26%	18%
BB	786	84%	64%	74%

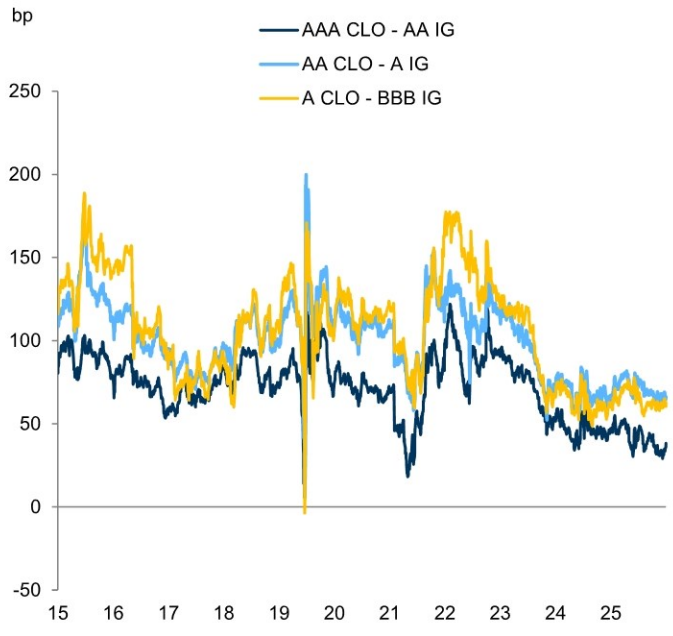
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 29: CLO new issue arbitrage spread
New issue CLO arbitrage spread, i.e., the difference between new issue loan spreads and new issue CLO spreads (across the capital stack)



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

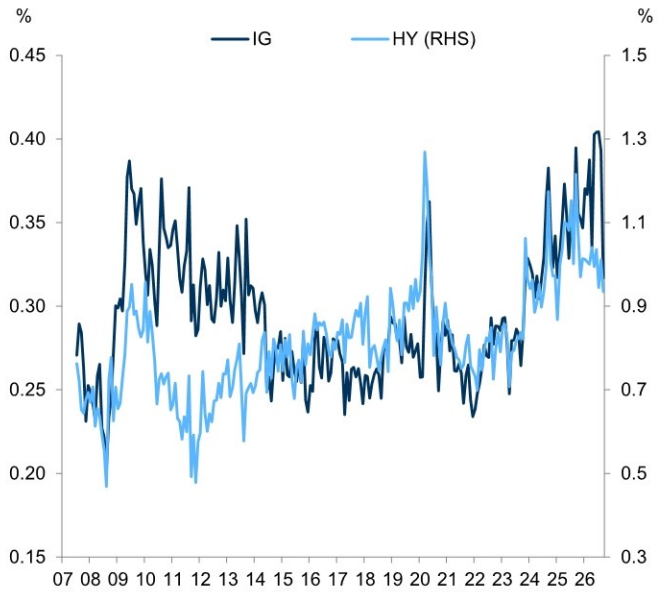
Exhibit 30: CLO vs. USD IG spread differential
CLO by rating and USD IG bond by rating spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

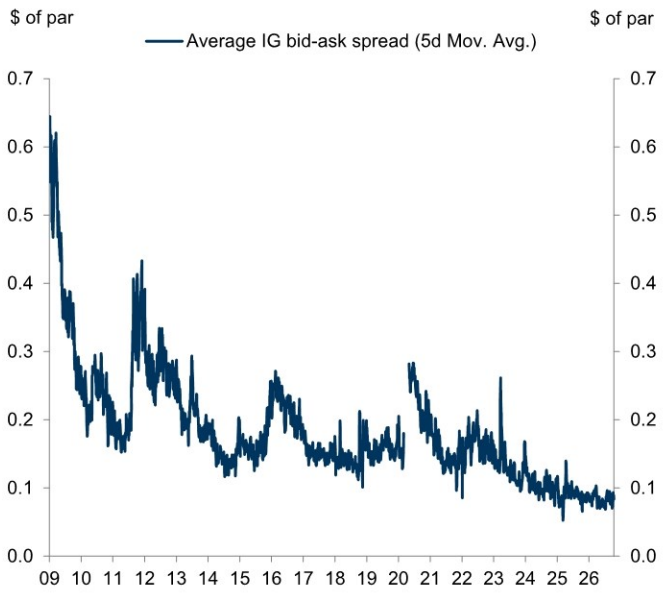
Microstructure dashboard: Liquidity conditions in the USD secondary bond market

Exhibit 31: Seasonally adjusted daily average turnover (trading volume/amount outstanding)



Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

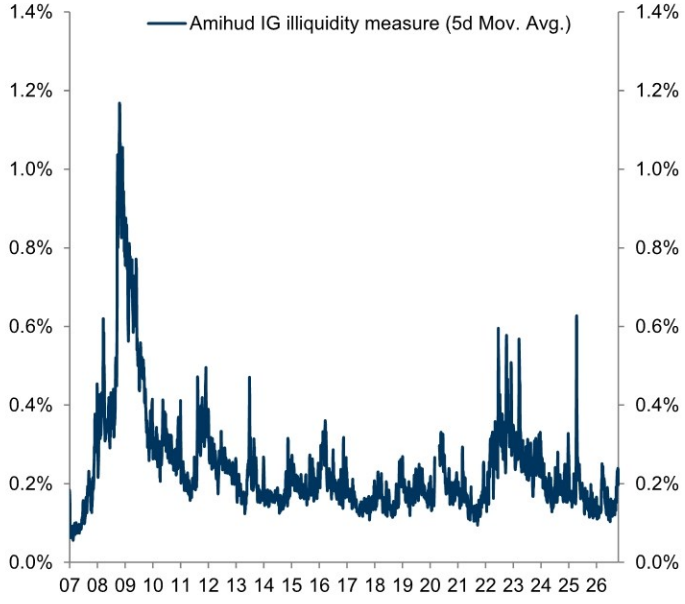
Exhibit 32: Average IG bid-ask spread
5-day moving average



Note: March and April 2020 are omitted.

Source: Trace, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 33: Amihud IG illiquidity measure
5-day moving average

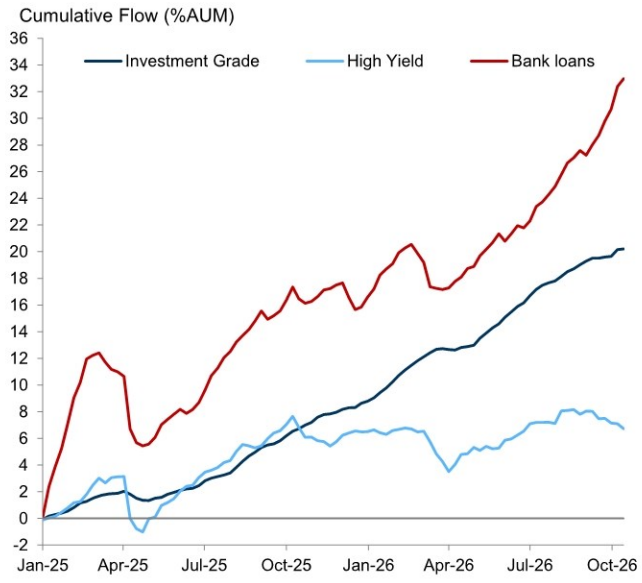


Note: March and April 2020 are omitted.

Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

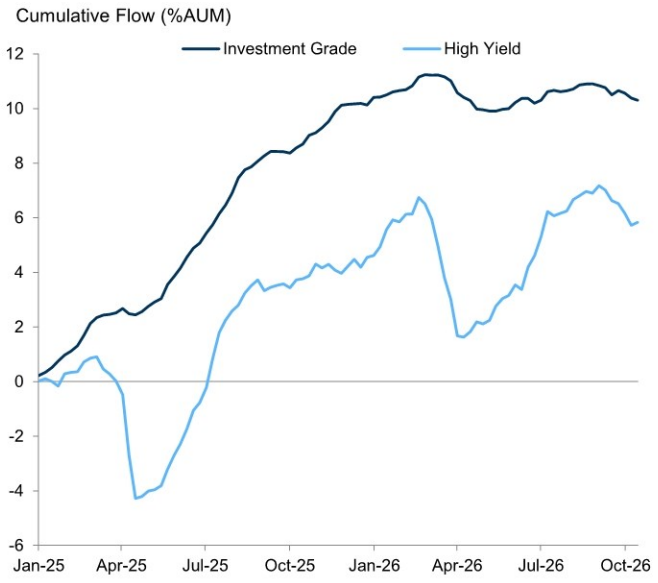
Flow dashboard: Mutual fund/ETF flows and foreign demand

Exhibit 34: Cumulative flows as a percentage of beginning of the year AUM for US credit funds



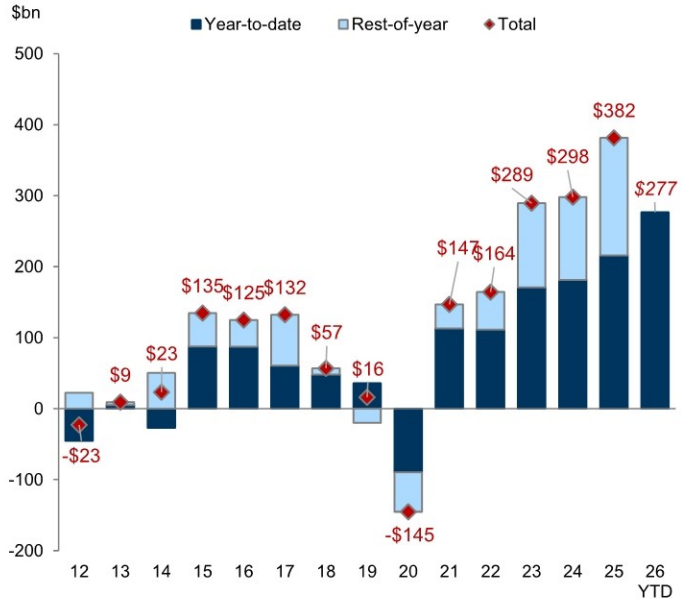
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 35: Cumulative flows as a percentage of beginning of the year AUM for European credit funds



Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 36: Net purchases of USD corporate bonds by foreign investors

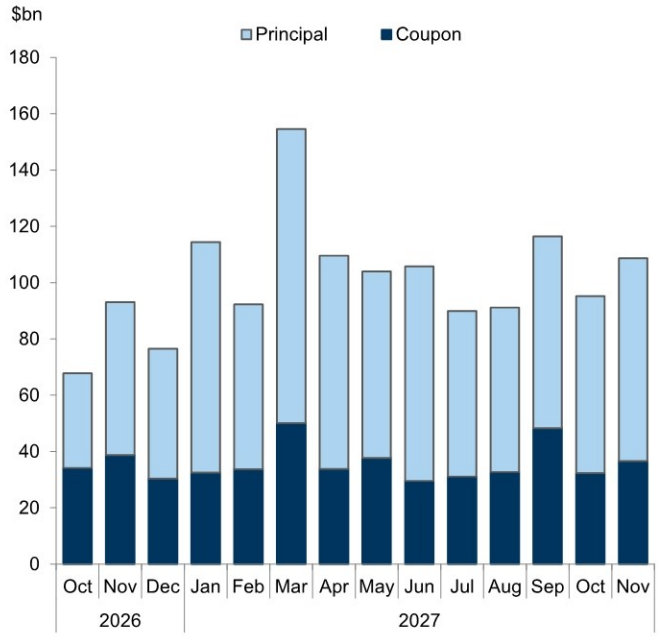


Note: As of July 2026.

Source: Department of Treasury, Goldman Sachs Global Investment Research

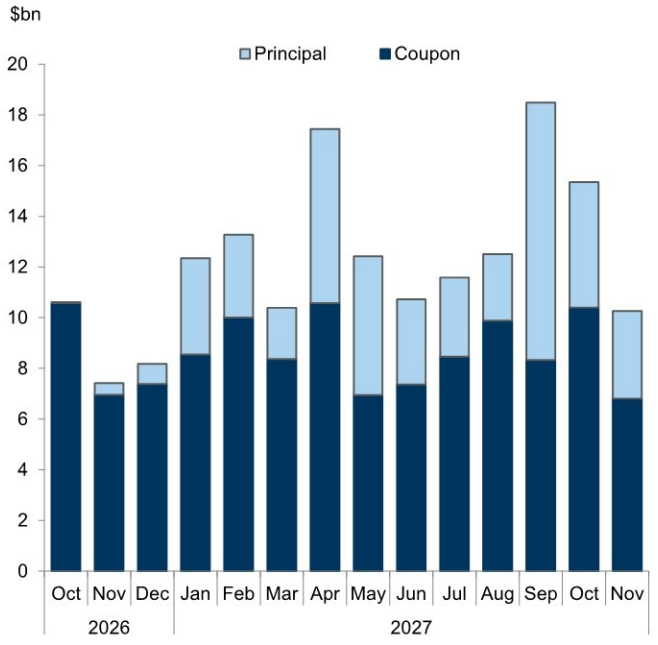
Flow dashboard: Upcoming coupon and principal payments

Exhibit 37: Our estimates of upcoming coupon and principal payments in the USD IG market



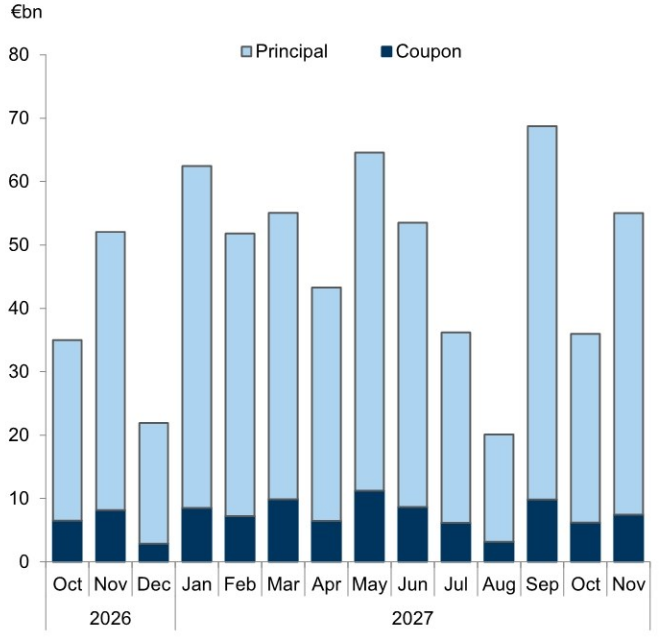
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 38: Our estimates of upcoming coupon and principal payments in the USD HY market



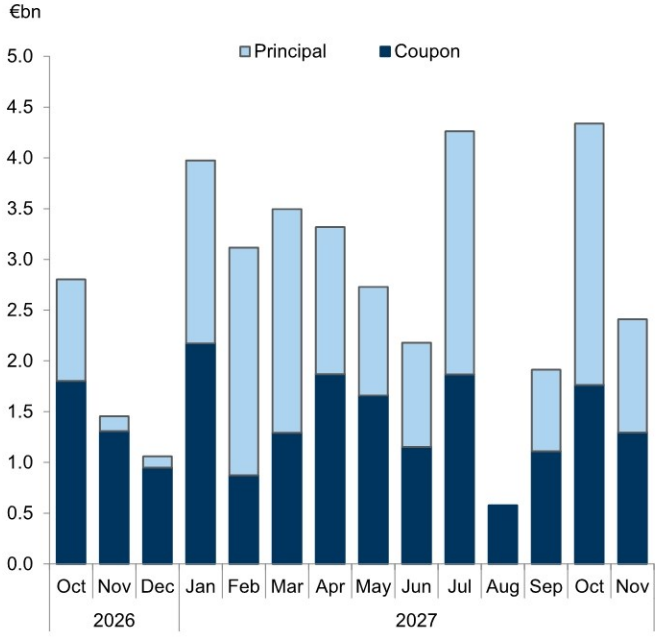
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 39: Our estimates of upcoming coupon and principal payments in the EUR IG market



Source: Bloomberg, Goldman Sachs Global Investment Research

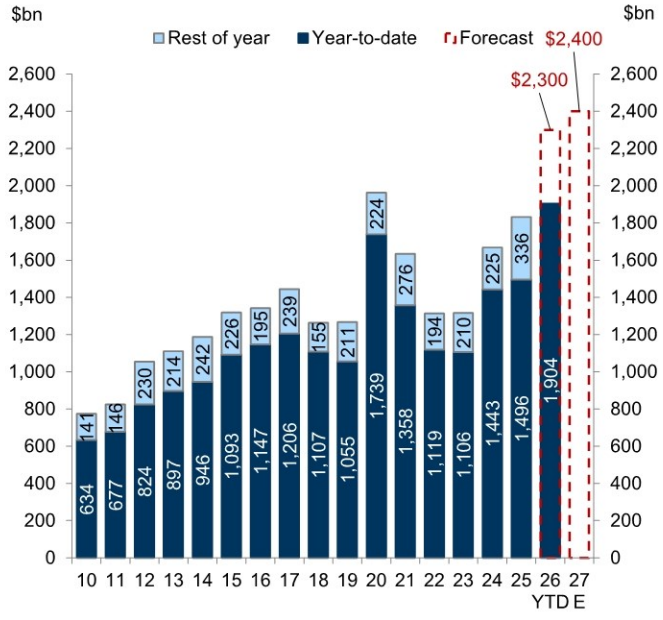
Exhibit 40: Our estimates of upcoming coupon and principal payments in the EUR HY market



Source: Bloomberg, Goldman Sachs Global Investment Research

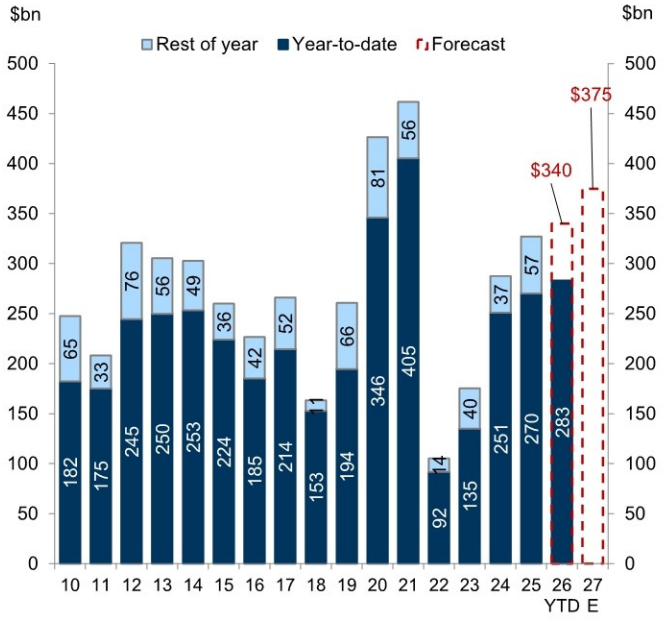
Primary market dashboard: USD gross issuance

Exhibit 41: USD IG gross issuance
Historical investment grade issuance: Same period year to date vs. rest of the year



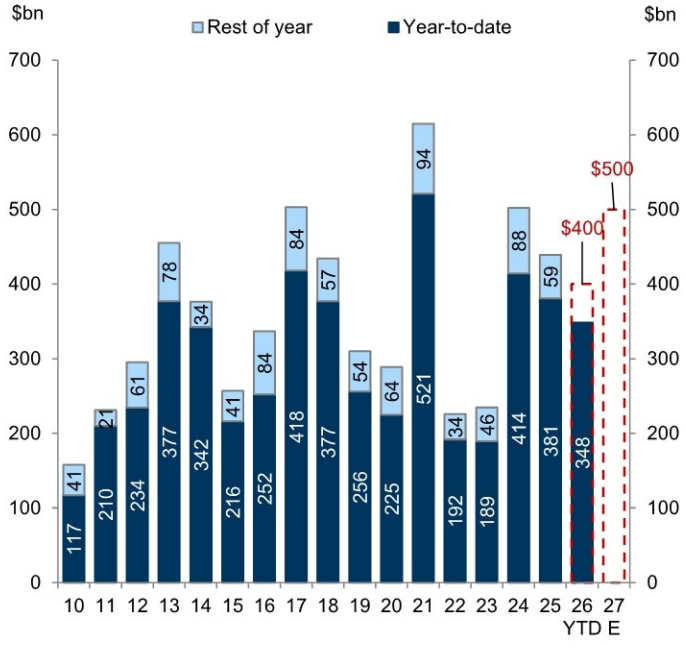
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 42: USD HY gross issuance
Historical USD high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

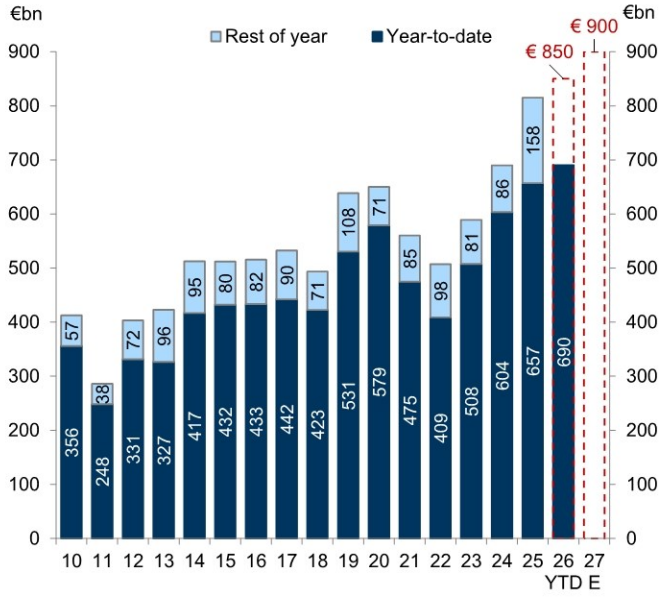
Exhibit 43: USD institutional leveraged loan gross issuance
Historical leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

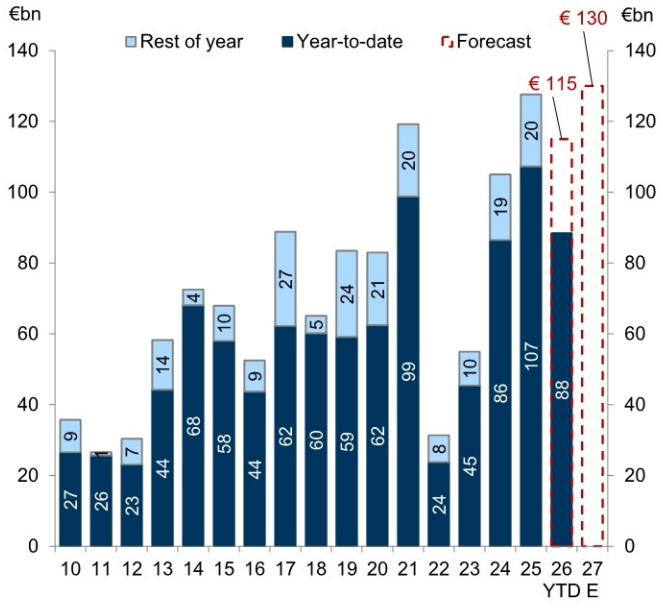
Primary market dashboard: EUR gross issuance

Exhibit 44: EUR IG gross issuance
Historical EUR investment grade issuance: Same period year to date vs. rest of the year



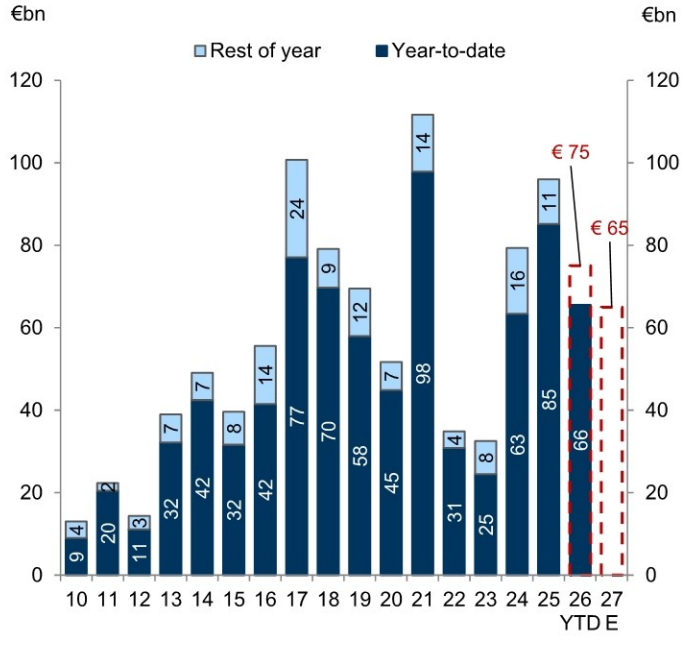
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 45: EUR HY gross issuance
Historical EUR high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

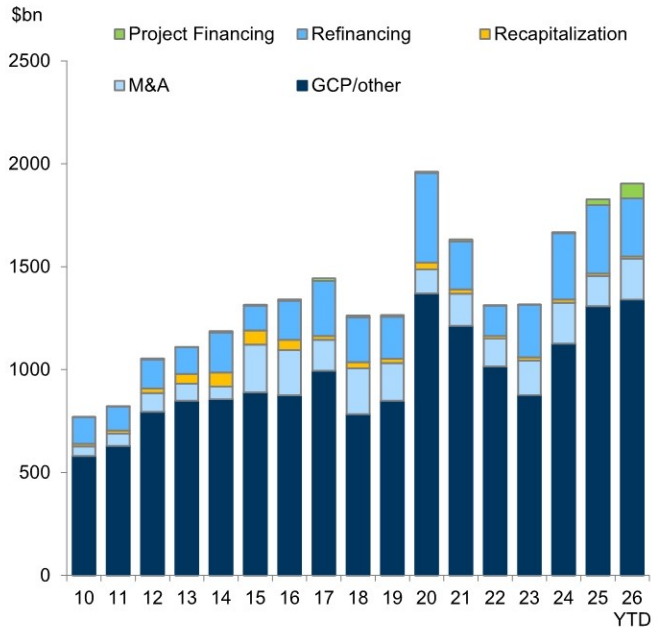
Exhibit 46: EUR Leveraged loan gross issuance
Historical EUR leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

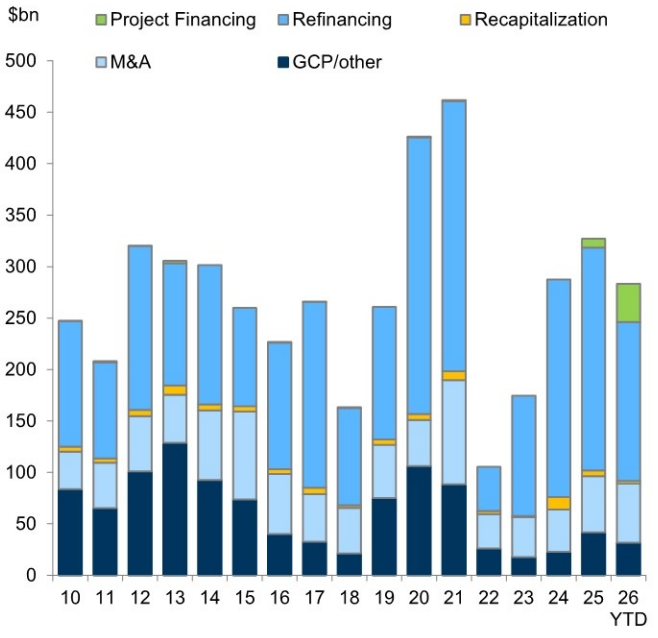
Primary market dashboard: Use of proceeds

Exhibit 47: Annual USD IG issuance volumes by use of proceeds



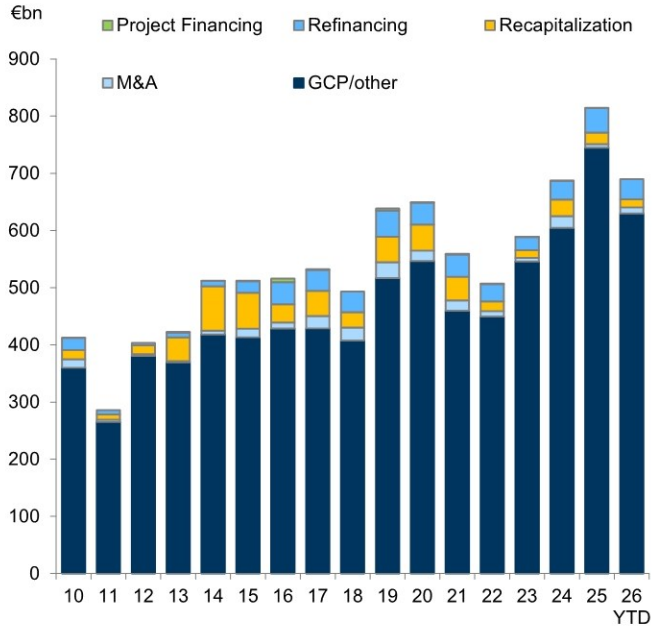
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 48: USD HY issuance volumes by use of proceeds



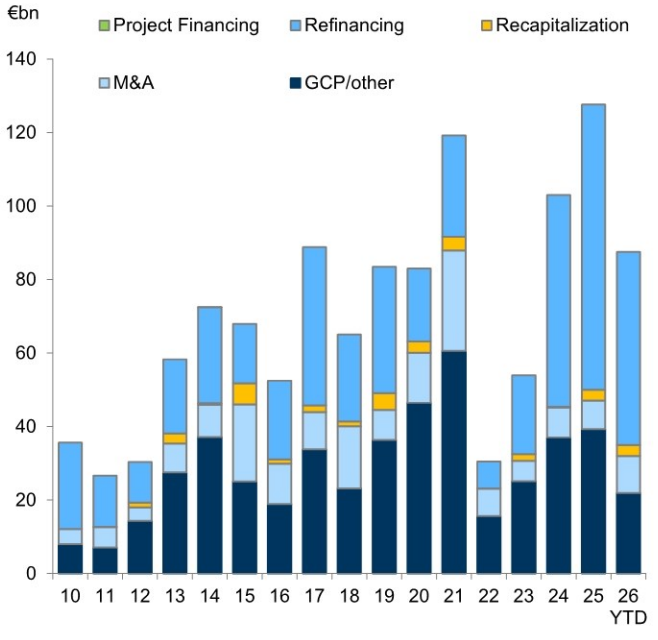
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 49: EUR IG issuance volumes by use of proceeds



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 50: EUR HY issuance volumes by use of proceeds

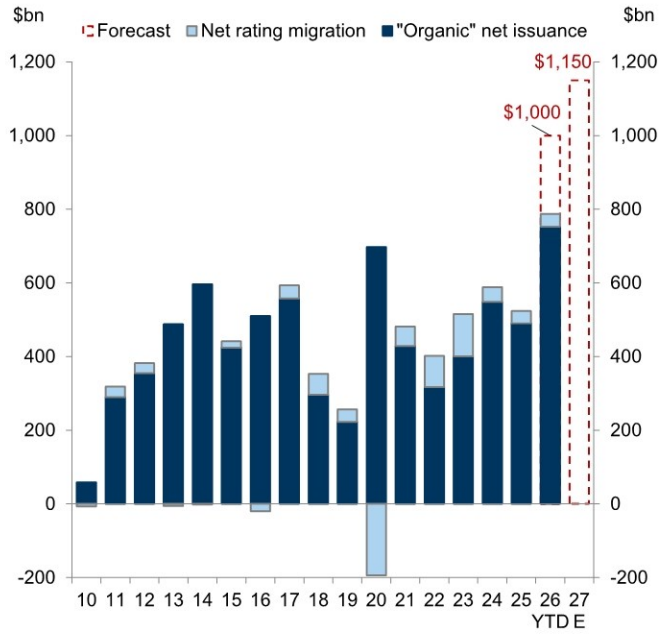


Source: Dealogic, Goldman Sachs Global Investment Research

Primary market dashboard: Net issuance

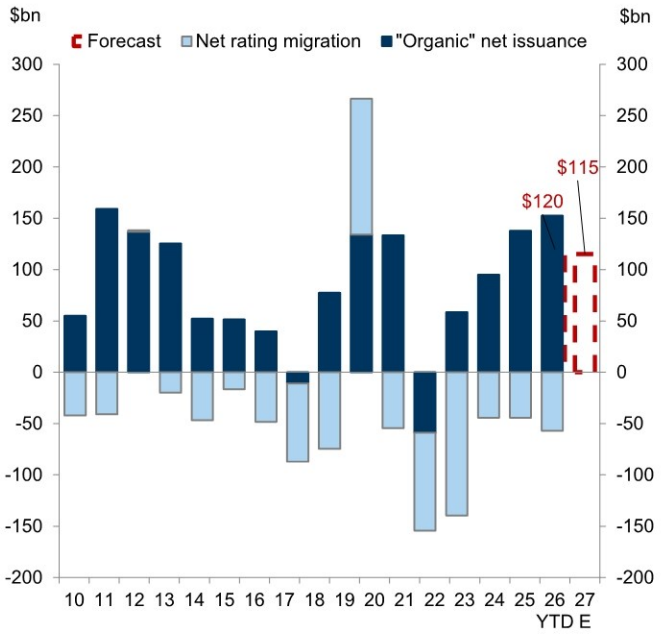
Note: Decomposition of the change in par value outstanding into net ratings migration vs. "organic" net issuance (gross issuance minus redemptions/matured bonds)

Exhibit 51: Annual USD IG net issuance



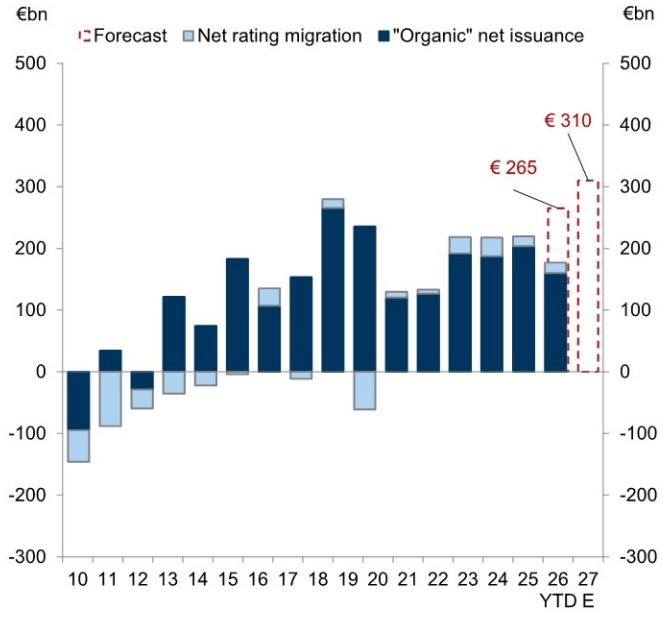
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 52: Annual USD HY net issuance



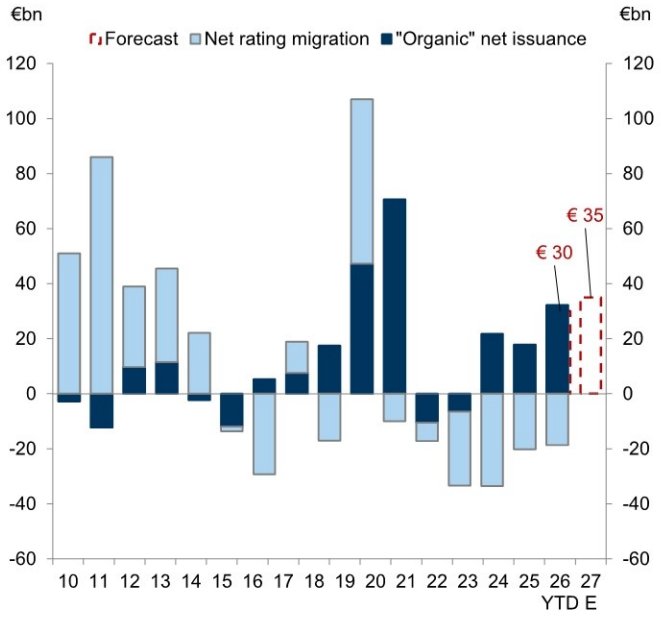
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 53: Annual EUR IG net issuance



Source: Bloomberg, Goldman Sachs Global Investment Research

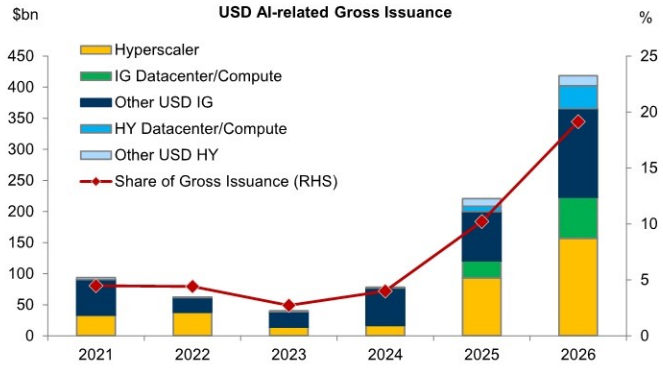
Exhibit 54: Annual EUR HY net issuance



Source: Bloomberg, Goldman Sachs Global Investment Research

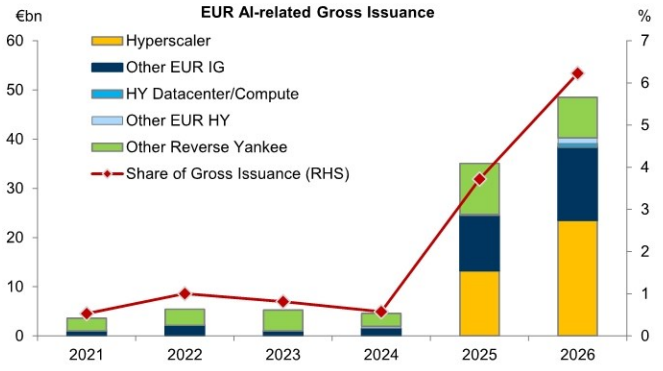
Primary market dashboard: AI-related issuance for the USD and EUR markets

Exhibit 55: USD AI-related Gross Issuance



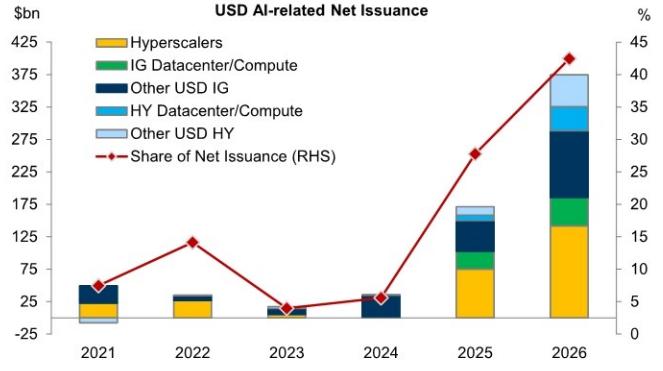
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 56: EUR AI-related Gross Issuance



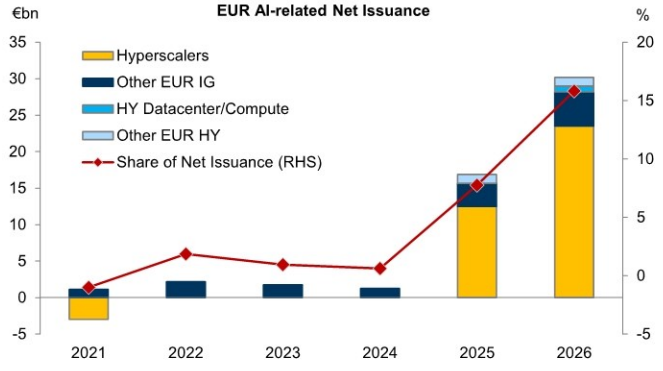
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 57: USD AI-related net issuance



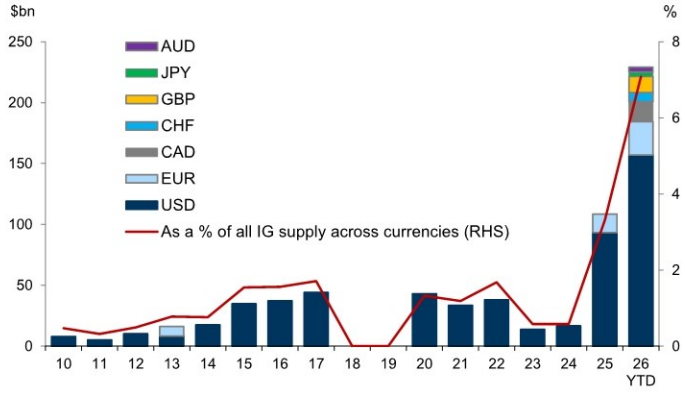
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 58: EUR AI-related net issuance



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 59: Hyperscaler debt issuance across currencies (in USD-equivalent)

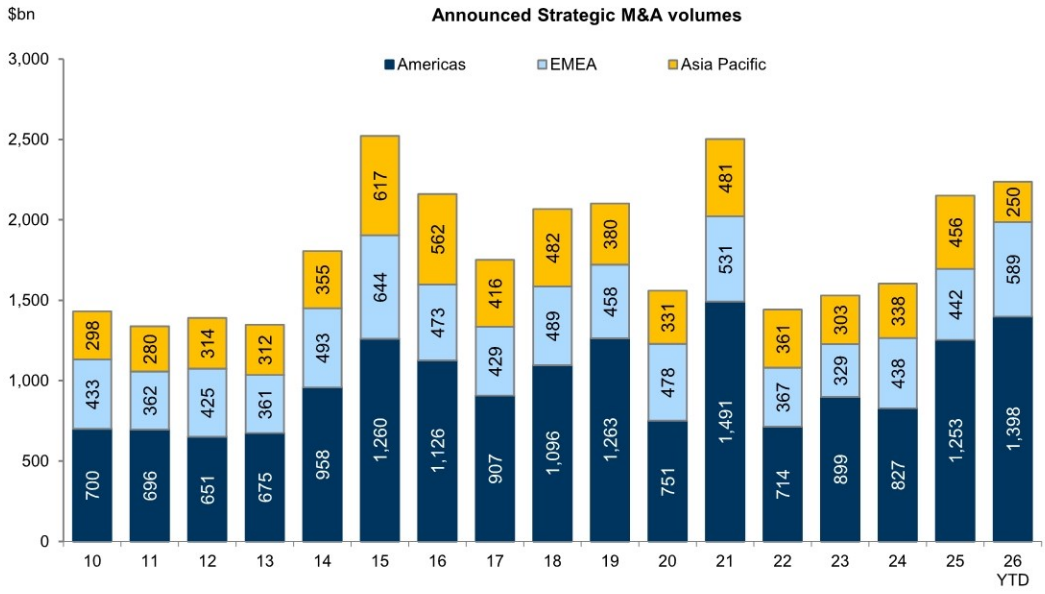


Note: Captures debt issued by the following companies: Alphabet (Google), Amazon, Microsoft, Meta, and Oracle.

Source: Dealogic, Goldman Sachs Global Investment Research

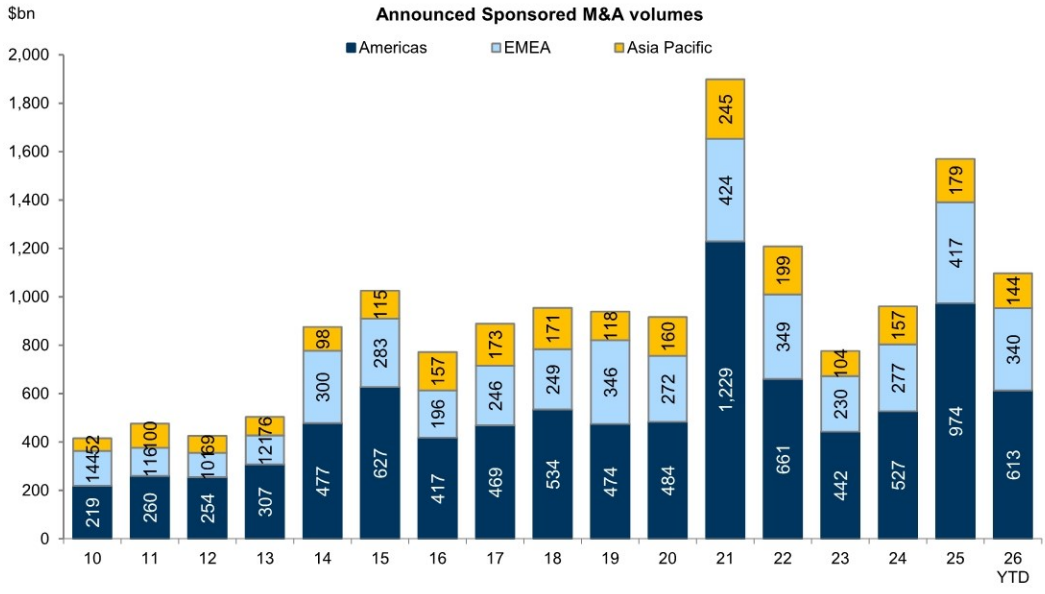
M&A volume

Exhibit 60: Annual strategic M&A volumes by region



Source: Dealogic, Goldman Sachs Global Investment Research

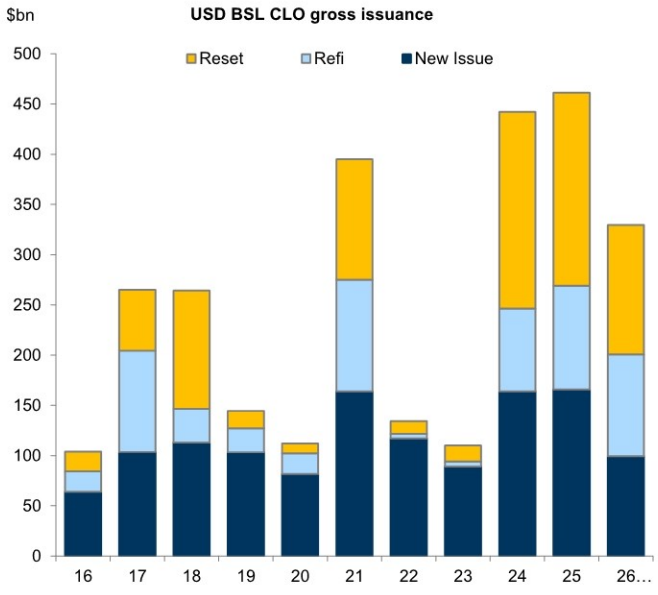
Exhibit 61: Annual sponsored M&A volumes by region



Source: Dealogic, Goldman Sachs Global Investment Research

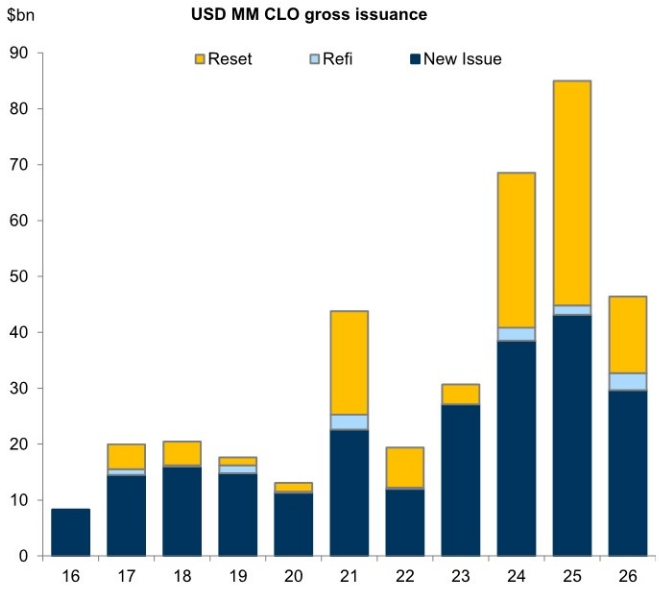
Primary market dashboard: CLO creation and refinancing

Exhibit 62: Annual USD broadly syndicated loan CLO creation



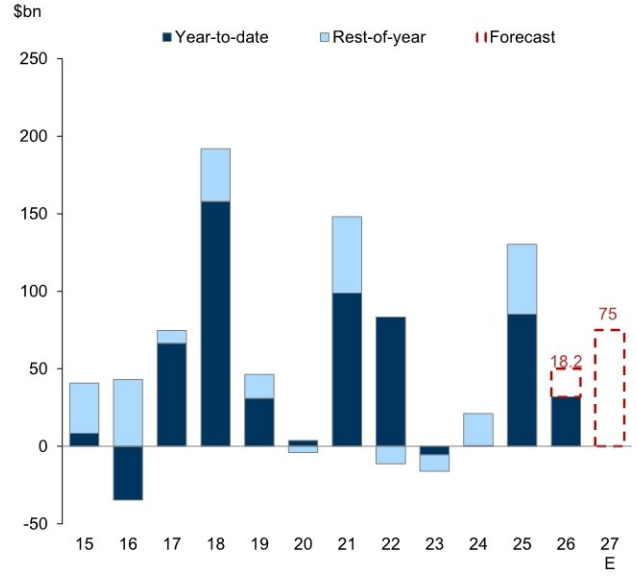
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 63: Annual USD middle market CLO creation



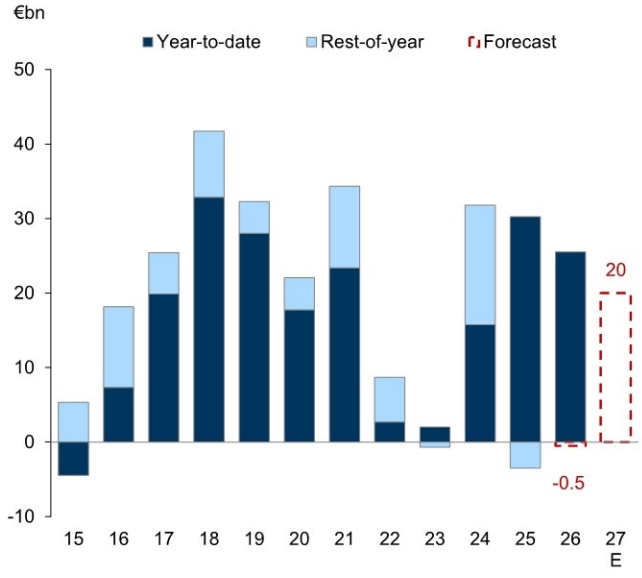
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 64: Annual USD leveraged loan net issuance



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

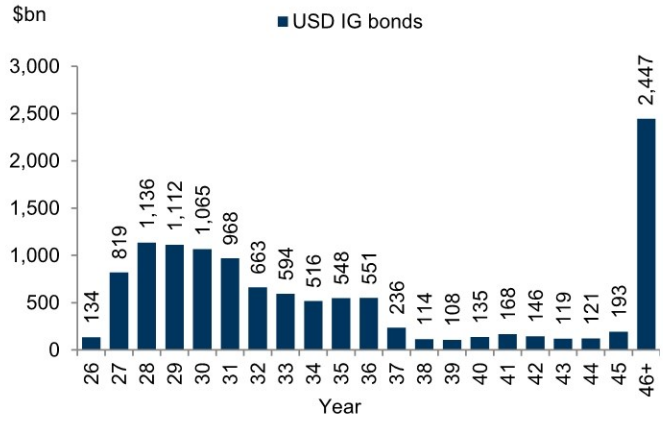
Exhibit 65: Annual EUR leveraged loan net issuance



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

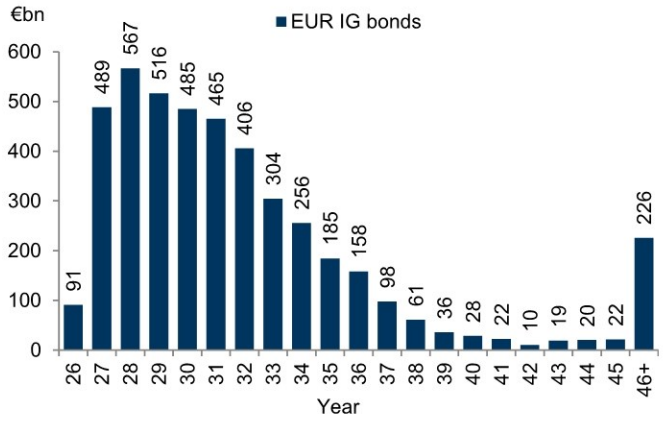
Primary market dashboard: Maturity walls

Exhibit 66: USD IG bond market maturity wall



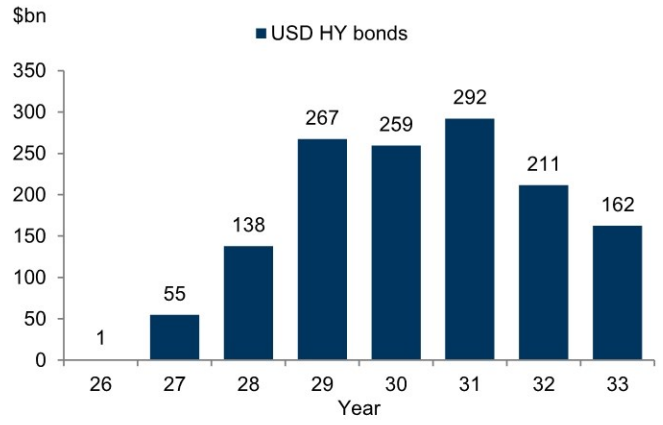
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 67: EUR IG bond market maturity wall



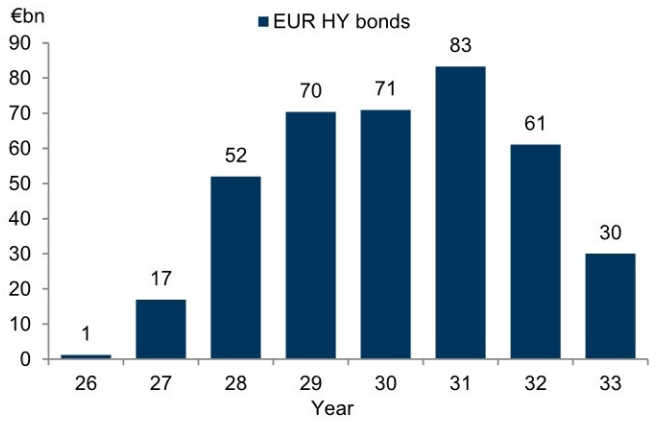
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 68: USD HY bond market maturity wall



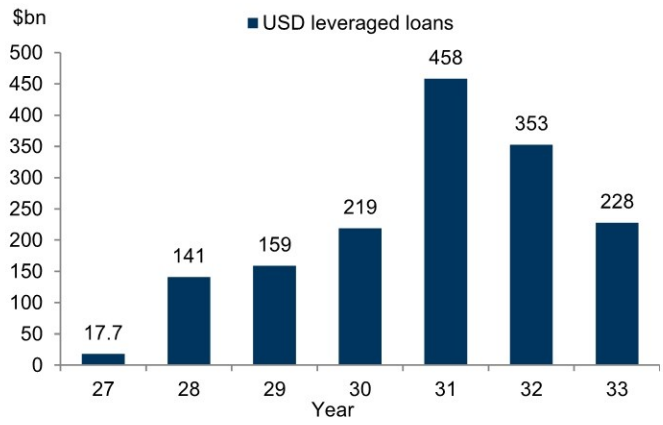
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 69: EUR HY bond market maturity wall



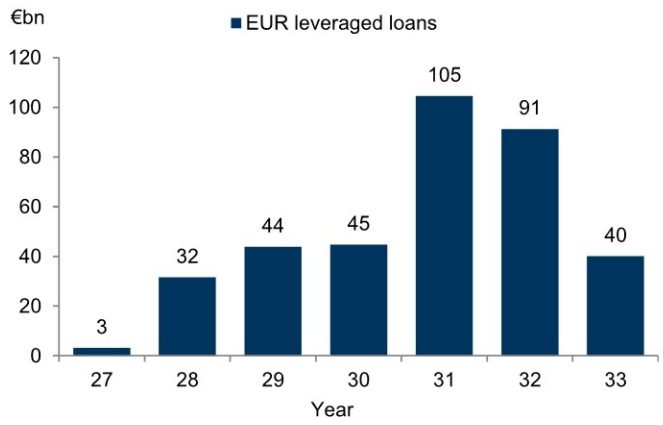
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 70: USD leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

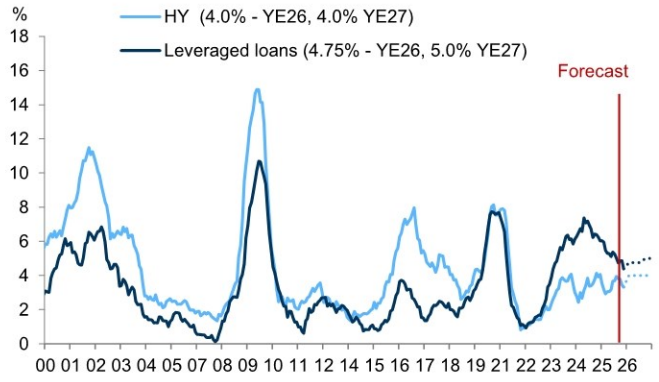
Exhibit 71: EUR leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

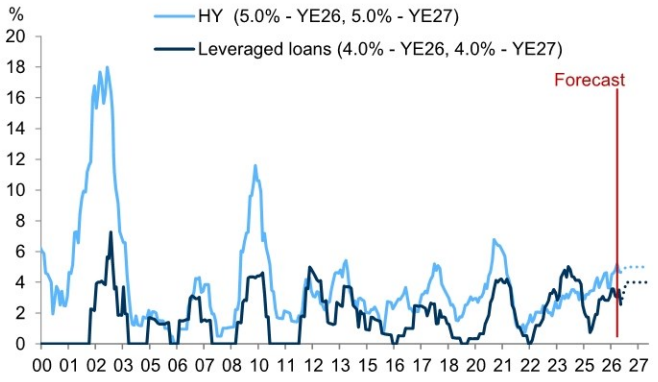
Default and ratings migration dashboard: Default trends, upgrades and downgrades

Exhibit 72: Default rates for US HY bond and leveraged loan issuers
12-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



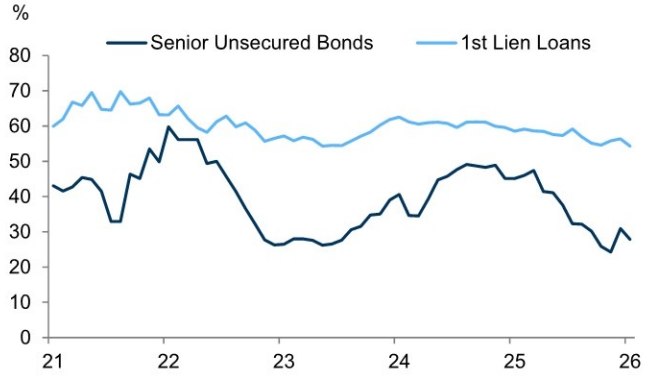
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 73: Default rates for European bond and leveraged loan issuers
12-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



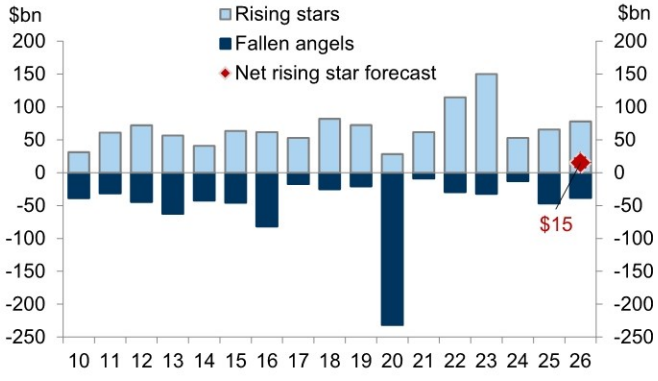
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 74: Annual recovery rates for US issuers



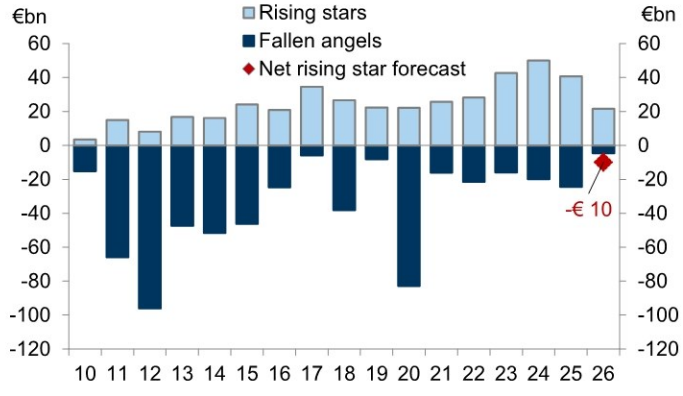
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 75: Fallen angels vs. rising stars in the USD market



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 76: Fallen angels vs. rising stars in the EUR market



Source: Bloomberg, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

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